HACKNITE CODE ROYALE 2026 • X402 & ALGORAND

Official Problem Brief

Agentic Payments (x402) on Algorand: Autonomous Micropayments

> *"Enable AI agents and services to transact autonomously through secure, verifiable, per-request micropayments on Algorand."*

FORMAT	THEME	DOMAIN
Hackathon Track	**Pay-Per-Use API Economy for AI Agents**	**Web3 & Autonomous AI Agents**

1. CHALLENGE OVERVIEW & CONTEXT

All projects must build and expose a working testnet x402 endpoint. This hackathon track focuses on Agentic Payments powered by x402 on Algorand, enabling AI agents, applications, and services to transact seamlessly via per-call micropayments instead of rigid subscription paywalls.

> **Core Challenge:** Traditional Web2 SaaS relies on credit cards, subscription paywalls, and manual API key signups—creating massive friction for autonomous AI agents that need to transact programmatically on demand.

Context & Ecosystem: As AI agents become primary economic actors on the web, machine-to-machine commerce requires machine-native payment rails. x402 turns HTTP 402 Payment Required into a cryptographic payment handshake on Algorand.

Benchmark & Opportunity: Projects can build single x402 capability endpoints or aggregators of agents for micropayments where multiple endpoints are atomically called as defined in the Entry Management Framework.

01	02	03	04	05	06	07	08
Client Request	402 Challenge	Client Signs	Auto Retry	Facilitator	USDC Settle	Tx Receipt	API Response

2. VALUE CHAIN SUB-TRACKS

DEV TOOLS & CODE AI

Developer & Code Intelligence Micro-APIs

"Pay-per-request developer utilities and code review endpoints."

Build microservices that charge developers or agents per request for code analysis, regex generation, and debugging.

- Roast My Commit Message API (Charge per request to review commit messages & suggest alternatives)
- Bug Reproduction Summarizer (Convert noisy bug reports into reproducible steps with severity scoring)
- API Changelog Explainer (Transform release notes into concise migration guides)
- Smart Regex Builder (Generate production-ready regex from plain English descriptions)
- Prompt Guardrail Checker (Detect prompt injection risks, jailbreaks, and policy violations)
- Git Diff Explainer (Explain code changes in simple language for founders & PMs)

ENTERPRISE & CONTENT MICRO-APIS

Business, Content & Sales Automation APIs

"Monetized AI endpoints for marketing, sales, and HR."

Expose specialized pay-per-call endpoints for lead generation, content creation, and document auditing.

- Cold Email Personalizer (Create personalized outreach emails from structured lead info)
- Ad Copy Variant Generator (Generate marketing copy optimized for different tones)
- Resume-to-Role Match Scorer (Score resumes against job descriptions & identify skill gaps)
- Contract Clause Risk Tagger (Highlight risky clauses & missing protections)
- Policy-to-Checklist Converter & Domain Name Idea Critic
- README-to-Pitch Deck Generator & Meme Caption Engine

AGENT INFRASTRUCTURE

Agent Governance, Trust & Guardrails

"Security, spend policies, and trust scoring for autonomous agents."

Build endpoints that provide financial boundaries and trust verification for AI agents.

- Agent Spend Policy Guard (Enforce spending limits & policies before autonomous agents execute payments)
- Trust Score API (Generate trust scores for wallets, services, or agents using historical behavior)
- Review Sentiment Watcher (Analyze customer reviews & identify complaint trends)
- Mini Fine-Tuned Support Classifier (Classify support tickets into billing, technical, refund, urgent)
- Meeting Notes to Action Items (Extract tasks, owners, and deadlines from transcripts)

ENTRY MANAGEMENT FRAMEWORK

Atomic Agent Aggregators & Multi-Step Workflows

"Atomically call multiple x402 endpoints in a single agent pipeline."

Build aggregator services that coordinate multiple paid x402 endpoints as defined in the Entry Management Framework.

- Multi-Step Research Agent (Coordinate multiple paid services to generate comprehensive research reports)
- Atomic multi-endpoint micropayment routing and settlement
- Unified transaction-linked receipt aggregation
- Scalable pay-per-workflow execution for complex multi-agent pipelines

3. PROVIDED DATASETS & APIS

RESOURCE NAME	DESCRIPTION & SCOPE

4. JUDGING CRITERIA & WEIGHTAGE

WEIGHT	CRITERIA	DESCRIPTION
30%	**x402 Protocol Flow**	Complete execution: 402 Challenge → Client Sign → Auto Retry → Facilitator Settle → Response receipt.
25%	**Real Pay-Per-Call Model**	Clearly identified paying user and real pay-per-call business model without subscriptions.
20%	**Technical Execution & Algorand**	Clean @x402 integration, USDC ASA settlement on Algorand, and transaction ID in response.
15%	**Innovation & Utility**	Practical utility for autonomous AI agents, applications, or services.
10%	**Documentation & Deployment**	Working testnet x402 endpoint, clean deployment docs, and mainnet-ready architecture.

5. VISION STATEMENT

> **"Enable AI agents, applications, and services to transact autonomously through secure, verifiable, per-request micropayments on Algorand."**
>
> Participants are encouraged to build production-oriented solutions where payments happen seamlessly between users, services, and autonomous agents using x402 on Algorand.